



Embedded systems power medical devices

contract and pump blood away from the heart and as the impulse moves away from each specific chamber of the heart, that section relaxes. Take a moment to understand the complexity of the above process. If medicine is not your thing then still try to understand that the process described is a very delicate one. If the natural pacemaker malfunctions leading to abnormal heartbeats, then blackouts, cardiac arrest, and even death is possible. So what do you do if the pacemaker is faulty? Replace it with an artificial electronic pacemaker. This pacemaker is an embedded device which uses electrical impulses which are delivered by electrodes contacting the heart muscles and so regulates the beating of the heart. A typical pacemaker consists of the computer part and the pacing wire. The computer part has the required electronic components and the battery for the power supply. Pacemakers are now pretty reliable and more than 2 million people in the US alone are living with a pacemaker.

But there is one dark side to having an awesome pacemaker. Recent pacemakers have an added wireless capability which can allow doctors to work on anything faulty in it without having to open it up. Only problem is that if someone has a laptop and has the skills to do it he can hack into that pacemaker and make it issue a lethal shock, effectively killing whoever it is housed in. What a nice way to turn such a beautiful life giving device into an agent of death.

Anti-hacker technologies

Embedded systems are helping create a stronghold of a defense for sensitive security systems. While previously embedded systems were used in the aid of security systems, they would create a single unit out of many processors and hence would centralise the processors and data banks, making them soft targets for hackers. Now what is happening is that even the smallest kernel of information has its own embedded system. That is almost like defending every note in your bank account with an A-grade security system, but not as visual. So in effect, for the hacker to gain access to these sensitive documents, he or she would have to break into each specialised and unique coded security apparatus one painful kernel at a time. You could almost feel sorry for these hackers. Almost.